CIM-Hub Carsharing Tanzania

Project Report

App-Supported Carsharing Service for Tanzania

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Table of content

1. General Information 3
1.1 Content Description of the Project 3
1.2 Project Framework 3
1.3 Team Members 4
2. Project Progress and Organisation 5
2.1 Report on Project Progress and Planning 5
2.1.1 Geography & Marketing Milestones 5
2.1.2 IT Milestones 5
2.1.3 Adjustments in the Current Semester 6
2.2 Hindering Factors in the Project 6
2.2.1 Planning and Execution 7
2.2.2 Workflows 7
2.2.3 Problem Resolution 7
2.3 Facilitating Factors in the Project 7
2.4 Handling of Knowledge & Knowledge Transfer 7
2.4.1 Interdisciplinary Working Method 8
2.4.2 Communication Formats 8
3. Reflection and Outlook 8
3.1 Reflection on Implementation and Decisions 8
3.2 Summary of Results 8
3.2.1 Backend Results 8
3.2.2 Mobile App Results 9
3.2.3 Key Insights 9
3.3 Future Work & Outlook 9
3.3.1 Immediate Technical Priorities 9
3.3.2 Strategic Priorities for the Coming Semester 9

1. General Information

1.1 Content Description of the Project

Subject of Investigation & Project Goal: The main goal of this project is to establish an app-supported carsharing service for Tanzanians.

Problem Statement: Due to bad infrastructure and a major lack of security in existing, self-organized carsharing methods, the project aims to provide an app-based solution for safe, reliable, and cost-effective car sharing, with a focus on long-distance travel.

Update for this semester: Since the previous semester, the project scope has shifted significantly. The Tanzanian partner organization hired a full-time developer and independently developed their own application. As a result, the focus of this semester's work moved away from app development toward **integration and collaboration.** This represents a meaningful expansion of the partnership and a positive step toward local ownership of the technology meanwhile challenging projects efforts in Marburgs application.

1.2 Project Framework

- **Project Initiators:** The project was initiated by the CIM-Hub, Jörg Brenneisen, Brighton Ammy, and Jackson Barnaby.
- **Supervision:** Supervised by Prof. Dr. Simone Strambach, Prof. Dr. Kathleen Otto, Prof. Dr. Michael Stephan, and Prof. Dr. Gabriele Taentzer.
- **Timeframe:** The project is in continuous development, with an estimated app launch no earlier than May 2026.
- **Update for this semester:** The overall timeframe remains unchanged, with the app launch estimated no earlier than May 2026. However, within the Tanzanian team: the hired full-time developer has requested additional time to ensure a stable and complete implementation, while the broader Tanzanian team is pushing for a faster

1.3 Team Members

Mohamed Fakhr(M.Sc. Business informatics):I managed communication with the geography sub-team and acted as the IT team representative during discussions with the Tanzanian stakeholders and developers. Additionally, I handled the technical development of the database.

Hikmat Samadov (M.Sc. Data Science): I focused on the backend architecture and DevOps workflow, aiming to make our system production-ready by migrating microservices to Kubernetes from Eureka and setting up the Google Cloud deployment pipeline. Beyond infrastructure, I standardized API contracts, introduced dynamic Swagger/OpenAPI documentation and implemented key features like the review service and enhanced authentication (password reset/change). I also added admin functionalities based on direct feedback from the Tanzanian team (admin back-end) and introduced a monitoring stack (loki, grafana, prometheus) and audit logging for admins to improve system visibility.

Sumeet Kumar (M.Sc. Computer Science): I was responsible for Flutter frontend develpment and REST API Migration from firebase, Designed and implemented client side architechtured using Clean Architechture principles with Riverpod for state managment, Helped in migration from the Firebase Authentication to a JWT-based REST API backend, removing all Firebase dependencies that were being used and replacing them with a Centralized Dio HTTP client that included token injection, error handling, and request interceptors. Debugged and resolved API endpoint routing issues by cross-referencing the Swagger/OpenAPI specification.

Ahmed Elmesery (M.Sc. Computer Science): Responsible for Flutter frontend development, feature implementation, and helped with the REST API migration and architecture design. Participated in designing and implementing the client-side architecture using Clean Architecture principles with Riverpod for state management. Worked on the migration from Firebase to a REST API backend. Implemented core user-facing features including the passenger trip search and booking flow, driver trip creation, and the management interface. Integrated OpenStreetMap Nominatim for real-time address autocomplete with debouncing and offline fallback. Designed the data model layer with code-generated JSON serialization and custom converters to match the backend's microservices API contract. Implemented a client-side fallback search strategy for cases where the primary server endpoint fails.

Lionel Schaefer (M.Sc. Economic Geography): This semester, the role encompassed geography and marketing management, along with coordination responsibilities. This included maintaining regular communication with the IT manager to align on technical developments, as well as coordinating with the Tanzanian team leader and organizing calls with the broader Tanzanian team. Additionally, CIM Hub professors were consulted to seek advice and keep supervision informed of project progress. On the marketing side, ideas were developed for both the survey distribution strategy and the upcoming app launch, contributing to the project's outreach.

2. Project Progress and Organisation

2.1 Report on Project Progress and Planning

2.1.1 Geography & Marketing Milestones

- **Survey:** A survey was conducted to gather market insights; however, participation was too low to draw meaningful conclusions. An alternative distribution approach was proposed but ultimately dismissed by the Tanzanian team as they did not see a necessity for further data collection at this stage.
- **Project Management Tool:** A Google Sheets-based project management tool was introduced to improve internal coordination, task tracking, and transparency across the group.
- **Stakeholder Communication:** Efforts were made to establish and maintain communication with key stakeholders, including Brighton, the broader Tanzanian team, the newly hired full-time developer, and supervising professors. Joint meetings involving both teams were organised to align on project goals and progress.
- **Marketing:** Initial exploration into social media marketing was undertaken as a potential channel for app distribution and outreach once the app approaches its launch phase.

2.1.2 IT Milestones

Completed milestones:

- Backend — Migration to Kubernetes & GCP: Successfully migrated from Eureka service discovery to Kubernetes DNS, deployed microservices to GKE Cluster, and

implemented a GitOps workflow with Artifact Registry. Java runtime upgraded from 17 to 21 (LTS). Implemented unified response models and dynamic Swagger/OpenAPI documentation.

- Mobile & Backend Integration: Integration is nearly complete. Most endpoints are functional, with only one recently developed endpoint remaining.
- REST API Integration for Authentication: Firebase has been fully replaced with GCP microservices backend via REST API.
- Location & Maps Integration: Comprehensive geolocation and mapping capabilities integrated using OpenStreetMap, flutter_map, and Nominatim geocoding. Includes real-time GPS tracking, dynamic route polylines, map markers for pickup/dropoff points, and Haversine distance calculations as a client-side fallback.

Planned but incomplete:

- M-Pesa Integration: Remains pending due to stakeholder coordination challenges. Frontend-backend connection achieved first successful integration but requires stabilisation.
- Driver Licence Verification: Currently any licence is accepted without verification; a dedicated verification flow is required for security purposes.

2.1.3 Adjustments in the Current Semester

- Migration from Netflix Eureka to Kubernetes-native DNS discovery, eliminating the Eureka registry server. The deployment model shifted from Docker Compose-only to Kubernetes-first (local Kind + GKE cloud), enabling self-healing, horizontal scaling, and rolling updates.
- Migration from Firebase cloud database to a REST API backend, improving performance and scalability. A central API manager handles all server communication, automatically managing authentication tokens and logging requests for debugging.
- Implementation of a complete mapping and location system across all platforms, including real-time address suggestions, GPS-based current location display, interactive maps with pickup/dropoff markers, automated distance calculations, offline map support via cached data, and in-app location permission dialogs.

2.2 Hindering Factors in the Project

2.2.1 Planning and Execution

Communication and coordination posed significant challenges throughout the semester. Since the beginning of the year, seven calls took place, including one joint meeting with the full Marburg and Tanzanian teams. Despite this, response rates remained a persistent issue, particularly in the final stages of the project. Following the joint meeting, in which commitments were made by both sides, communication difficulties continued: Edwin remained unresponsive to the IT team's technical inquiries, while Brighton's availability was limited due to travel and other commitments — resulting in missed deadlines and prolonged periods of no response. These bottlenecks have notably slowed progress in the critical final phase of the project.

2.2.2 Workflows

Despite the challenges of coordinating a diverse, cross-functional group, communication within the team was maintained throughout the semester. Active efforts were made to align sub-group managers and keep priorities on track, even as differing schedules and goals made full-group meetings difficult to arrange consistently. The introduction of a project management tool reflected a proactive step toward improving internal transparency and coordination.

2.2.3 Problem Resolution

To address these challenges, communication was increasingly channelled through project management to ensure a more structured flow of information and accountability. The support of CIM Hub professors was sought to help mediate and maintain momentum, particularly during periods of low engagement from both internal and external stakeholders. A project management tool was introduced to improve coordination, task tracking, and transparency.

2.3 Facilitating Factors in the Project

- Planning and Execution: Internal planning and execution proceeded very smoothly this semester, driven by clear technical milestones
- Organization:Weekly IT subgroup meetings with structured agendas improved task visibility. Using Discord for scheduled meetings and WhatsApp for async coordination balanced formal documentation with rapid problem-solving.

2.4 Handling of Knowledge & Knowledge Transfer

2.4.1 Interdisciplinary Working Method

In contrast to the previous semester, where a lack of internal communication caused sub-groups to work mostly in isolation, communication issues were fully resolved this semester. A reliable meeting routine was established: the IT team held weekly meetings to coordinate software development and architecture, while bi-weekly meetings with the geography team addressed empirical data and overall project alignment. Day-to-day coordination and quick questions were handled via a shared WhatsApp group. Communication with the Tanzanian stakeholders was also substantially improved.

2.4.2 Communication Formats

- Discord: Used for scheduled weekly and bi-weekly team meetings
- WhatsApp: Group chat for day-to-day coordination and fast async replies

3. Reflection and Outlook

3.1 Reflection on Implementation and Decisions

While the microservice architecture offers significant long-term benefits, the initial setup and learning curve proved more time-consuming than anticipated. A simpler approach might have been a better fit for the current project stage, as it would have allowed the team to maintain focus on feature work rather than infrastructure concerns.

3.2 Summary of Results

3.2.1 Backend Results

- Working microservices architecture on Kubernetes (GKE)
- 6 core services operational: API Gateway, Auth Service, User Service, Trip Service, Review Service
- Java 21 runtime, MongoDB Atlas persistence
- Comprehensive OpenAPI/Swagger documentation
- GitOps deployment pipeline
- First successful mobile-backend integration

3.2.2 Mobile App Results

- Flutter-based application with complete UI for passenger and driver flows
- Features: authentication, trip search, booking, payment selection, trip management, map integration
- Successfully migrated from Firebase temporary backend to REST API connection

3.2.3 Key Insights

This semester reinforced the importance of centralised communication and clear commitments in cross-functional group projects. Key insights include:

- Goal alignment and transparency across all stakeholders are critical for smooth coordination and steady progress
- Integration with the Tanzanian team's newly developed app has not yet been completed; the practical impact of this semester's efforts for the Tanzanian side therefore remains limited for the time being

3.3 Future Work & Outlook

3.3.1 Immediate Technical Priorities

- Stabilise mobile-backend integration — complete API contract alignment and support additional back-end routes
- M-Pesa integration — coordinate with Tanzanian stakeholders for provider testing and sandbox access
- Review System Completion — connect the mutual rating system to the ride completion flow
- Testing & QA — implement comprehensive unit/integration tests and automated integration tests for common user flows
- Security Hardening — review authentication/authorisation flows and secure environment variables

3.3.2 Strategic Priorities for the Coming Semester

The immediate priority for the coming semester is a structured discussion among all key stakeholders to determine the project's direction — particularly regarding the ongoing communication challenges and the question of which application to move forward with. Specific focus areas include:

- Consolidating efforts around a single application to avoid duplication and ensure all parties are aligned on one shared solution
- Supporting marketing efforts in preparation for the app launch, helping to build visibility and reach for the product
- Onboarding new participants to contribute to marketing and outreach once the technical direction is clarified